

CATAMARCA, Argentina

WMO: 872220

Lat: 28.5936S

Lon: 65.7553W

Elev: 1490

StdP: 13.92

Time Zone: -3.00 (W03)

Period: 96-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
6	35.4	38.5	17.6	14.1	57.1	21.3	16.8	55.2	32.1	63.1	28.1	62.4	4.3	50	0.519

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	19.5	101.4	71.8	98.7	71.3	96.4	70.9	76.0	93.0	74.8	91.6	73.8	90.0	16.4	50

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
71.5	122.9	81.9	70.2	117.4	81.6	69.2	113.4	81.4	40.7	93.2	39.6	92.0	38.6	90.7	82.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 34.7	31.5	27.9	DB	28.8	106.5	3.8	2.1	26.1	108.1	23.9	109.3	21.8	110.5	19.1	112.1	(4)
(5)			WB	26.7	79.2	3.4	2.1	24.2	80.7	22.3	81.9	20.4	83.1	17.9	84.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	70.5	82.8	80.5	76.7	69.8	62.1	55.1	55.0	61.1	68.2	75.0	78.8	81.6	(6)	
	DBStd	11.84	5.88	6.57	6.28	6.76	6.39	5.38	5.97	7.18	8.17	7.59	7.11	6.38	(7)	
	HDD50	45	0	0	0	0	2	15	22	5	1	0	0	0	(8)	
	HDD65	1014	0	1	3	30	134	298	313	161	59	13	2	0	(9)	
	CDD50	7531	1018	855	828	595	376	168	177	350	546	774	864	980	(10)	
	CDD65	3029	553	436	367	175	43	1	4	41	154	322	417	516	(11)	
	CDH74	34245	6848	4954	3402	1360	317	32	91	633	1698	3520	4950	6440	(12)	
	CDH80	16671	3667	2479	1457	443	55	2	20	222	698	1684	2510	3434	(13)	
(14) Wind	WSAvg	11.4	13.7	12.9	11.7	9.4	8.6	6.6	7.7	9.8	12.6	14.2	15.1	14.7	(14)	
(15) Precipitation	PrecAvg	15.4	3.0	2.8	1.9	1.1	0.5	0.4	0.2	0.1	0.2	0.9	1.7	2.6	(15)	
	PrecMax	25.0	7.0	6.8	3.7	4.6	2.0	1.6	1.0	0.9	1.6	3.4	4.5	5.2	(16)	
	PrecMin	9.0	0.7	0.7	0.2	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.1	0.1	(17)	
	PrecStd	4.9	1.5	1.6	1.1	1.1	0.5	0.4	0.2	0.2	0.3	0.8	1.1	1.0	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	103.8	102.3	97.8	93.0	85.2	77.9	82.0	93.8	98.2	101.4	103.0	104.6	(19)	
		MCWB	73.7	73.5	73.1	70.4	66.3	61.0	61.4	64.4	66.8	68.9	70.2	71.7	(20)	
	2%	DB	100.2	98.4	94.2	88.4	80.8	73.4	75.6	86.1	91.7	96.8	98.4	100.5	(21)	
		MCWB	72.6	72.9	72.3	68.5	64.2	58.3	57.4	61.3	63.8	68.0	69.8	71.6	(22)	
	5%	DB	97.1	95.1	91.2	85.3	77.4	70.1	72.1	80.9	87.7	93.0	95.2	97.1	(23)	
		MCWB	72.3	72.5	71.6	67.6	62.5	55.9	55.7	59.3	62.6	67.0	68.7	71.2	(24)	
	10%	DB	94.3	92.0	87.8	82.1	74.1	67.4	68.6	76.4	83.6	89.3	92.1	94.0	(25)	
		MCWB	72.0	72.1	70.4	66.6	61.2	54.8	53.7	57.1	60.4	65.1	67.5	70.5	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	77.8	77.2	76.5	73.8	68.1	63.1	63.0	65.6	69.3	73.8	74.2	76.4	(27)	
		MCDB	95.6	93.9	91.9	86.3	79.5	73.4	79.0	88.3	91.5	93.6	94.3	95.3	(28)	
	2%	WB	75.7	75.7	74.7	71.9	65.9	60.3	60.1	63.1	66.4	71.0	72.2	74.7	(29)	
		MCDB	93.2	92.1	89.2	82.0	77.3	70.1	72.6	83.0	87.4	89.8	91.6	93.1	(30)	
	5%	WB	74.6	74.5	73.3	70.2	64.4	58.2	57.3	60.5	64.4	68.8	70.8	73.5	(31)	
		MCDB	91.5	90.4	86.5	80.3	74.8	67.2	69.1	78.7	83.4	87.5	89.5	91.7	(32)	
	10%	WB	73.5	73.4	71.9	68.4	62.7	56.2	54.9	58.0	62.0	67.3	69.6	72.3	(33)	
		MCDB	89.7	88.5	84.2	78.9	72.0	64.9	66.5	74.3	80.0	84.6	87.6	89.4	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	19.5	18.7	18.2	18.7	19.9	22.4	24.7	25.3	23.8	21.5	20.4	20.2	(35)	
		MCDBR	21.9	21.4	21.6	23.5	24.0	27.4	30.6	33.5	29.1	26.5	23.7	23.2	(36)	
	5% WB	MCWBR	6.8	6.4	7.4	8.9	10.7	14.7	15.9	15.4	11.5	9.1	8.0	7.2	(37)	
		MCWBR	19.9	19.5	19.1	19.2	21.1	23.7	27.4	31.8	26.2	23.3	21.2	20.7	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.401	0.394	0.375	0.361	0.327	0.314	0.311	0.350	0.400	0.409	0.399	0.400	(40)	
		taud	2.409	2.430	2.482	2.466	2.510	2.516	2.498	2.373	2.235	2.289	2.356	2.397	(41)	
	Ebn at Noon		299	295	291	279	275	272	278	278	278	287	297	300	(42)	
		Edh at Noon	40	38	34	31	27	26	27	34	43	43	42	41	(43)	
(44) All-Sky Solar Radiation	RadAvg		2195	1970	1673	1349	1092	1024	1158	1472	1811	2052	2244	2261	(44)	
	RadStd		130	130	120	118	107	86	69	76	120	154	149	112	(45)	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	+1.78	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (2 neighbors)	N/A	N/A	N/A	+1.21	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page